

MACHINE LEARNING (S403011)

Fall 2025

Contacts:

- Lecturer: [Prof. Sebastian Engelke](#)
- Teaching assistants: [Christophe Valvason](#), [Manuel Hentschel](#)

Schedule:

- Course: Tuesday 16:15 - 18:00, UniMail M R030
- Seminar: Thursday 16:15 - 18:00, SCIII 1S081 - Edouard Kellenberger

Objectives:

Upon success the student will be able to:

- Choose appropriate machine learning methods among the ones enumerated in the content.
- Perform the corresponding statistical analysis using the programming language Python.
- Discuss advantages and drawbacks of each method.
- Discuss mathematical issues associated to each method.

Description: This course provides the background necessary for a competent application and interpretation of machine learning approaches supporting business analytical decision processes. It first overviews the most broadly used methods for supervised learning in regression and classification tasks. It then discusses the problem of model selection and how it can be performed with methods such as cross-validation. The course then overviews more recent methodologies related to flexible classification, bagging and boosting, support vector machines and tree-based methods. Modern techniques such as deep neural networks and convolution based image classification will also be studied. After these courses, students should be able to evaluate with a critical mindset the findings of machine learning methods, and to assess their applicability, as well as their advantages and limits in applications. Moreover, students should possess a good understanding both of the underlying statistical ideas and the main theoretical properties of these methods.

Grading Policy:

- Semester project (30%).
- Final written exam, 2h, closed book (70%). Allowed material:
 - A non-programmable calculator.

Same format and rules for the retake exam.

- In case of an excused absence at the final exam, the semester project will count 30% at the retake.
- In case of an unexcused absence or failed attempt of the final exam, the student can choose whether to keep the grade of the semester project or have the retake exam count for 100% of the final grade.

Tentative Course Outline:

Week	Topic	Events
1	Introduction + Motivation	
2	Regression Framework + Linear Regression + Gradient Descent	
3	Training/Test error + Cross-validation	
4	Lasso/Ridge + Classification Framework	
5	LDA/QDA + Logistic Regression	
6	Image recognition + Decision trees	
7	Bagging + Random Forest	
	Reading week	No classes
8	Boosting + Maximal Margin Classifier	
9	Support Vector Classifier + Support Vector Machine	
10	Neural Networks + Deep Neural Networks	
11	Back-propagation + Regularization of Neural Networks	
12	CNN + further topics	
13	Q&A	